

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 27

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• About 40-50 traps • Because the running mean stops changing / fluctuating with fewer than this the mean is unreliable	2
01.2	• Widely scattered results (large standard deviation) mean a single sample may not be representative • More samples are needed so that the effect of variability is averaged out and the mean is representative / significance can be assessed	2
01.3	<i>Any 3 from:</i> • Same type and size of pitfall trap, set for the same length of time • Traps emptied at the same time of day / same weather conditions in both habitats • Same number of traps and same method of positioning (random or systematic) in each habitat • Other variables that could affect results (eg rainfall, temperature) monitored or controlled	3
01.4	• Flying insect activity changes with weather over minutes to weeks (temperature, wind) and diurnally • Samples on different occasions are needed to produce a mean that represents the typical value or to identify trends • The interval between samples should match the rate at which the factor changes, found from a preliminary study	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	

Question	Marking guidance	Marks
02.1	<i>Any 4 from:</i> • Zooxanthellae are photosynthetic algae living inside the polyp cells • The algae absorb CO₂ and excretory wastes from the polyp for photosynthesis and growth (efficient nutrient recycling) • They supply the polyp with oxygen and glucose - over 95% of its food energy • In return the algae gain a protected habitat and nutrients they could not absorb from the nutrient-deficient water • Polyps also feed on plankton captured by cilia and nematocysts	4
02.2	• Corals deposit a skeleton of calcium carbonate (calcite) • Each carbonate molecule requires the absorption of a molecule of carbon dioxide, so reefs are a carbon sink	2
02.3	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Importance: cover 1% of ocean area but support over a quarter of marine species; fisheries (fish, lobsters, conch); medicines (AZT, Ara-C, dolastatin, secosteroids, bryostatin, trabectedin, bone grafts); climate control as carbon sink; erosion protection of low-lying islands; tourism income • Threats: physical damage (anchors, divers, litter, coral extraction), collection of ornaments (CITES Appendix II), sedimentation from rivers and coastal development, pollution (oil, nutrients causing algal shading, sewage, pesticides, sunscreen), ocean acidification, bleaching from warming, overfishing (giant triton / crown of thorns starfish), introduced species (red lionfish), loss of mangroves and seagrass nurseries • Conservation: marine protected areas / national parks (Great Barrier Reef, Belize, Chagos), mooring buoys, diver education, No Take Zones which can increase catches elsewhere, sustainable ecotourism and fishing, protection of mangroves and seagrass, erosion and pollution control, planning control of ports and resorts • Fishing controls: closed seasons, minimum and maximum catchable sizes, bans on dynamite, trawling and small-mesh nets • Evaluation: local measures cannot control global threats (warming, acidification) which need international action; enforcement in remote areas is difficult; conflicts with local livelihoods; large vs small NTZs • Justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
03.1	<i>Any 2 from:</i> • Potato bean (N America): high protein content • Morama bean (southern Africa): protein content similar to soya, drought resistant • Yeheb tree (Somalia): edible nuts, drought resistant, grows on poor soils • Perennial maize (Mexico) / Kernza: no need to buy seed each year, less ploughing so less soil erosion • American bison: graze grassland too poor for cattle • Common eland: browse and graze so can be kept with cattle without severe competition	2
03.2	• Indigenous species are better adapted to local conditions such as climate, pests / diseases and soil • So they have higher survival rates / productivity is not reduced, as it is when introduced species are kept in areas to which they are not adapted	2
03.3	• They grow in soils with a high salt content, so their genes could be used in breeding programmes to increase the salt tolerance of crops grown on soils salinised by irrigation	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11; AQA spec 3.1.2.1.1.	
04.1	• Diversity: HEP (wet winters), solar (summer) and wind peak at different times so supply is more reliable - diverse systems resist change • No single source could meet demand all year; battery storage on the island grid smooths fluctuations • Calm, dry, dull periods still leave gaps so diesel generators provide security (fossil fuel use minimised, not yet eliminated)	3
04.2	<i>Any 3 from:</i> • Community ownership (Isle of Eigg Trust with Scottish Wildlife Trust) managing the island for people and wildlife • Wood biomass heating; insulation such as straw-bale housing; some electric vehicles • Habitat protection and creation: fencing rare plants from grazing, planting indigenous woodland (rowan, birch, willow); eagles and hen harriers nesting • Recyclables shipped off; garden / food waste composted; low-impact food production; beach litter collection	3
04.3	• Small groups can organise, agree and act quickly; decisions benefit people who see the results directly • Resources and wastes are local and visible so feedback is immediate, unlike cities supplied from far away	2
04.4	• Renewable energy is abundant but geographically and temporally variable; a continental grid moves electricity from areas of surplus (desert solar, offshore wind) to areas of demand • DC grids do not need every generator synchronised in phase as AC grids do, so coordination across many countries is easier	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 384-389; AQA spec 3.6.4.	
05.1	• Its mass	1

Question	Marking guidance	Marks
05.2	- The 24-hour rotation period means the surface is not exposed to sunlight for long, so temperature extremes / excessive heating and cooling are reduced - The axis is tilted relative to the orbit so the amount of insolation at each latitude changes through the year, producing seasonal variations in conditions	2
05.3	- The distance from the Sun gives surface temperatures between 0 and 35 °C in most areas: too close and water would boil / evaporate, too far and it would freeze - Atmospheric pressure (produced by the mass of gases held by gravity) is high enough to stop the water boiling / evaporating away; at much lower pressure there would be no liquid water	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	
06.1	- EDGE: Evolutionarily Distinct and Globally Endangered - few close relatives, often the last species of their genus / family (Bactrian camel, pygmy hippo, northern bald ibis); loss would remove a unique branch of evolution / genes - Endemic: found in only one area (often islands - Galapagos, Madagascar, Seychelles: Aldabra giant tortoise, red ruffed lemur); if lost locally there is no population elsewhere to recolonise - Keystone: disproportionately important for the structure of the community relative to its abundance - predation of dominant species, seed dispersal, habitat creation (jaguar, forest elephants keeping paths and water holes open, wolves, beavers); losing them causes loss of many other species - Flagship: high public profile (tiger, panda, elephant, orang-utan) raising funds and support that also protects the whole habitat and less charismatic species (fungi, bats, moths)	8
06.2	- A fragmented population may consist of isolated groups each too small to be viable (resources, inbreeding) even if total numbers are large - A species confined to a single area (eg bonobos in one country) is vulnerable to any local change / conflict, unlike chimpanzees spread across many countries	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 28-29; AQA spec 3.1.2.3.1.	
07.1	<i>Any 2 from:</i> - Hydrological cycle - Carbon cycle - Nitrogen cycle - Atmospheric circulation - Thermohaline circulation - Photosynthesis	2
07.2	- Enzymes reduce the activation energy of reactions - So reactions such as photosynthesis, decomposition and nitrogen fixation do not need high temperatures	2
07.3	<i>Any 1 from:</i> - Haber process (nitrogen fixation) - Incineration of wastes (decomposition) - Many manufacturing processes	1

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	
08.1	• Images / maps of large areas; land-use change, deforestation, flooding • Vegetation density and surface temperature • Wind velocity, sea-surface altitude, wave height, oil pollution	3
08.2	• Microwaves pass through clouds (unlike visible light)	1
08.3	• Very large / remote areas can be surveyed quickly and repeatedly so changes over time are detected • No need for researchers to visit inaccessible areas, reducing cost and disturbance; data are standardised across the whole area	2
08.4	<i>Any 4 from:</i> • Motion-sensitive cameras placed along trails record animals passing, using visible light by day and infrared at night • Each tiger has a unique pattern of facial / body stripes • Images are compared with the database so individuals are recognised rather than counted twice • The number of different individuals recorded gives a minimum population; repeated captures give territory size, movements, lifespans and social groupings • Limitation: only works for species with unique visible features	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-403, 406-407; AQA spec 3.1.2.3.6.	
09.1	• Rotation: each crop has its own pests; growing a different crop each year (4-5 year cycle) means pests left at the end of a season die before that crop returns, so populations cannot build up early • Beetle banks / hedgerows provide habitat and overwintering sites for natural predators (ground beetles, ladybirds) that eat aphids • Rotation also evens out nutrient demands and includes legumes	4
09.2	• Integrated control gives almost the same yield as chemical control (18.1 vs 18.5) with far fewer applications (1-2 vs 6) • It retains most of the biodiversity of cultural control (26 vs 31 species) whereas chemical control alone leaves only 8 • Cultural control alone loses about 12% yield; integrated control combines cultural methods, resistant varieties and carefully timed specific non-persistent pesticides only when essential - best balance of yield, cost and impact	3
09.3	• Large numbers of males are sterilised with gamma radiation and released • Females of many insects mate only once and store sperm, so mating with a sterile male produces no offspring; enough sterile males reduce fertile matings below replacement level (screw worm fly, Mediterranean fruit fly) • Only the target species' females mate with them so non-target species are unaffected; fails if sterilised males behave abnormally or re-colonisation occurs	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 314-318; AQA spec 3.5.1.1.	
10.1	• By-catch = $5.9 + 1.9 = 7.8$ t; total = 8.4 t • $7.8 / 8.4 \times 100 = 92.9\%$ (accept 93%)	2
10.2	• The nets have a very small mesh size so few non-target animals can escape	1

Question	Marking guidance	Marks
10.3	• They breed at an old age (skate about 11 years; Greenland shark 100-150 years) and produce few young • They may reach a catchable size before they are sexually mature, so fishing can remove the whole breeding population before it has bred • Populations recover slowly, unlike r-selected species (herring, mackerel) which breed young and produce many young; they are caught as by-catch in demersal trawls so fishing continues even though they have no commercial value	3
10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Impacts: population decline of k-selected species, by-catch (immature target fish, over-quota fish, non-commercial species; drift nets - whales, dolphins, turtles; longlines - albatrosses; pair trawls - porpoises), ghost fishing, seabed damage by demersal trawls, coral and seagrass damage, dynamite fishing, food-web effects • Catch quotas: limit weight landed; work well for single-species pelagic shoals but in mixed fisheries over-quota fish are discarded dead so the quota fails to protect them • Gear design: mesh size and square-mesh escape panels let small fish escape (diagonal mesh closes under drag); turtle exclusion devices; acoustic pingers for dolphins; curved hooks, decoys, sinkers and night setting to protect albatrosses; biodegradable / radio-tracked traps to reduce ghost fishing • Restricted effort: limits on boat size, engine power and days at sea; decommissioning; restricted methods - drift-net bans, demersal trawling bans on sensitive seabeds (Rockall Bank, Hatton Bank) • Evaluation: enforcement at sea is difficult; some methods reduce catch or raise costs; quotas need reliable stock data; effectiveness depends on species (single vs mixed shoals) and on international cooperation as fish migrate; as local stocks decline fishing moves to more distant waters • Justified conclusion weighing effectiveness of different methods	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-345; AQA spec 3.5.2.2.	

Question	Marking guidance	Marks
11.1	Discuss how human activities influence biodiversity, using examples from a range of different contexts. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Direct exploitation: food (fisheries, bushmeat), fashion, entertainment / pets, furniture and ornaments, traditional medicines, other products (whale oil); deliberate eradication of predators, competitors, vectors, pests • Changes in abiotic factors: water availability (drainage, abstraction, HEP), light (forest clearance), oxygen (organic pollution, thermal), nutrients (eutrophication), pH (acid mine drainage, acid rain), temperature (thermal pollution, climate change), turbidity, physical damage; inactivity (stopping plagioclimax management) • Changes in biotic factors: introduced competitors, predators, pathogens, hybridisation; loss of pollinators, seed dispersers, keystone species; food chain impacts (sea otters, puffins, jellyfish) • Habitat destruction: deforestation, farmland expansion, urbanisation, mining, reservoirs; fragmentation and edge effects • Contexts: temperate woodland, rainforest, coral reefs, oceanic islands, mangroves, Antarctica; agriculture, fishing, forestry • Positive influences: habitat creation, conservation management, reintroductions, legislation • Evaluation of which influences are most damaging and hardest to reverse; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26, 328-330, 341-344, 367-369; AQA spec 3.1.2.2, 3.5.1.3, 3.5.2.2, 3.5.3.3.	25

Question	Marking guidance	Marks
11.2	Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Definition of sustainability (Brundtland); population x per capita impact; dynamic equilibria, feedback, tipping points, diversity • Principles of natural systems: renewable low-density energy, low temperatures, material cycles, non-toxic wastes; contrast with linear human systems • Circular economy principles: cycling materials, renewable energy, supporting ecosystems, separation of technical and biological materials, diversity, connected systems, design for extended use and re-use, optimum production, new technologies • Application to land use, water, minerals, waste, pollution, energy, food with examples (Kalundborg, Eigg, Ol Pejeta, Rio Bravo, EU initiatives, DC grids) • Measuring progress: ecological footprint, biocapacity, global hectares, Earth Overshoot Day, carbon footprint, Living Planet Index • Barriers: economics, competition, consumer behaviour, need for international agreement, existing infrastructure • Judgement on the extent achievable Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-389; AQA spec 3.6.	25